KERALA POLYTECHNIC · REVISION 2021

Basic Architectural Drawing

Complete student lesson for course code **2181**.

Revision 2021 Semester 2 Engineering Science 4 modules

Course snapshot

Scheme: 3 (L:1, T:0, P:2)

Credits: No Credit

Contact: 45 periods per semester

Module hours listed: 41

Official Source Declaration

This lesson uses the supplied official syllabus PDF as the authoritative source for course information, revision, modules, topics, objectives, Course Outcomes, assessment information, official activities, practical requirements, and references.

Source handling: Official wording and numerical values are retained wherever supplied. Educational explanations, Malayalam support, examples, diagrams, and revision questions are added as study support and are not presented as additional official requirements.

Transparency note: Official assessment information is reproduced below from the supplied syllabus. Any limitation in the supplied PDF is not silently corrected.

Course Dashboard

Course code

2181

Semester

2

Credits

No Credit

Teaching scheme

3 (L:1, T:0, P:2)

Module-hour and outcome mapping

No.	Module	Official hours	Relevant CO / level
1	Graphic Representation and Lines	6	CO1
2	Graphical Language, Visual Composition and Graphic Conventions	12	CO2
3	Free-Hand Sketching of Indoor and Outdoor Spaces	15	CO3
4	Colour Theory and Visual Composition	8	CO4

Prerequisites

- Nil.

How to study this lesson

1. Read the module introduction and learning goals.
2. Open every topic and reproduce its example, sequence, or procedure.
3. Use Malayalam support as a bridge; retain English technical terms for examinations.
4. Attempt the module questions without looking at the answers.

Objectives, Course Outcomes and CO–PO Information

Official course objectives

1. To introduce the students to fundamental techniques of Visual representation.
2. To introduce the basic techniques of Drawing, Painting and Drafting.
3. To comprehend the basics of graphic design and three-dimensional composition.

Course Outcomes

1. CO1: Discuss basic principles of graphic representation. (6 hours; Understanding)
2. CO2: Develop a graphical language of architecture. (12 hours; Applying)
3. CO3: Develop the ability to make free hand sketches of indoor and outdoor spaces. (15 hours; Applying)
4. CO4: Identify colour schemes and their application in a visual composition. (8 hours; Applying)

CO-PO mapping as supplied

CO1: PO1 = 3.
CO2: PO1 = 3 and PO7 = 3.
CO3: PO1 = 3 and PO7 = 2.
CO4: PO1 = 3 and PO7 = 2.
Legend supplied in PDF: 3 = Strongly mapped, 2 = Moderately mapped, 1 = Weakly mapped.

Complete Syllabus Coverage

Use this checklist to confirm that every official module topic is represented in the lesson.

Complete official topic coverage checklist

Module	Topic code	Topic	Hours
module-1	M1.01	Classify types of lines	1
module-1	M1.02	Characteristics and visual impacts of lines	1
module-1	M1.03	Practice exercises with lines	2
module-1	M1.04	Represent design principles with lines	2

Module	Topic code	Topic	Hours
module-2	M2.01	Graphical language of architecture	2
module-2	M2.02	Visual compositions using points, lines, shapes, tone and texture	4
module-2	M2.03	Applying shades on varied shapes	2
module-2	M2.04	Lettering, symbols and scale	4
module-2	M2.ST1	Series Test I	2
module-3	M3.01	Free-hand sketches	1
module-3	M3.02	Types of three-dimensional views	3
module-3	M3.03	Free-hand drawing for visual and architectural representation	5
module-3	M3.04	Still life, interiors and exteriors in pencil	6
module-4	M4.01	Chromatic values, colour wheel and colour theory	1
module-4	M4.02	Primary, secondary and tertiary colours; prepare a colour wheel	1
module-4	M4.03	Colour schemes: harmony, contrast and degree of chromatism	6
module-4	M4.ST2	Series Test II	2

Learning Roadmap

1. Graphic Representation and Lines
CO1 · 6 hours

2. Graphical Language, Visual Composition and Graphic Conventions
CO2 · 12 hours

3. Free-Hand Sketching of Indoor and Outdoor Spaces
CO3 · 15 hours

4. Colour Theory and Visual Composition
CO4 · 8 hours

The roadmap follows the order of the supplied syllabus.

OFFICIAL MODULE 1

Graphic Representation and Lines

Graphic representation begins with the controlled use of lines. Students learn to classify lines, observe their visual effects, practise them repeatedly, and use them to communicate rhythm, harmony, character, balance, emphasis, scale, and proportion.

Hours: 6

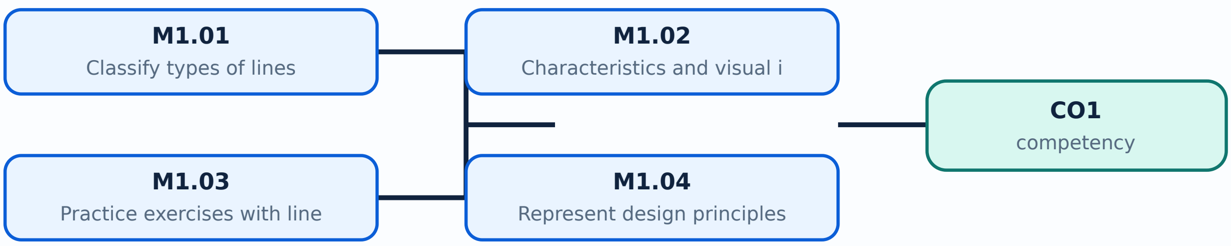
CO1 · Bloom level: Understanding

Learning goals

- Classify the line types named in the syllabus and describe their visual impact.
- Practise consistent line quality and controlled line exercises.
- Use lines to express design principles and interpret scale and proportion.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-1: the listed topics build the module competency.

M1.01 — Classify types of lines (1 hours)

Concept and explanation

Lines are the primary marks used in visual and architectural representation. The syllabus names vertical, horizontal, diagonal, zig-zag, curvilinear, and spiral lines. Each type has a directional and expressive character: vertical lines can suggest height or stability, horizontal lines calmness, diagonal lines movement, and curvilinear lines softness or continuity. Classification should be based on geometry and visual behaviour, not on decoration alone.

> **Malayalam support:** Line types □□□□□□□□□□□□ direction-□□ visual character-□□ □□□□□□□□. Vertical line □□□□/□□□□□□□ □□□□□□□□□□□□□□; horizontal line □□□□□□□□□□ diagonal line movement-□□ □□□□□□□□□□. Technical drawing-□ line name English-□ □□□□□ □□□□□□□□□.

Study points

- Keep the line families visually distinct; practise with constant pressure; label each exercise; do not confuse a line type with the design principle it may express.

> **Engineering / workplace application:** Architectural plans, elevations, furniture sketches, circulation diagrams, and presentation boards depend on a controlled line vocabulary.

> **Illustrative example:** Prepare six labelled strips: vertical, horizontal, diagonal, zig-zag, curvilinear, and spiral. Add one sentence describing the visual effect of each.

> **Common mistake:** Changing line weight randomly makes a drawing difficult to read; a classification exercise should show the characteristic of each line clearly.

M1.02 — Characteristics and visual impacts of lines (1 hours)

Concept and explanation

The visual impact of a line depends on direction, length, thickness, continuity, spacing, and repetition. A heavy continuous line may establish an edge, while a thin or broken line may indicate a secondary or hidden condition. In composition, repeated lines can create rhythm; related line qualities can create harmony; a contrasting line can create emphasis. These are visual relationships, so the student should observe them in actual sheets and sketches.

Malayalam support: Line-□□□□ direction, thickness, continuity, spacing □□□□□□ visual impact □□□□□□□□□□. Repetition rhythm □□□□□□□□□□□; related qualities harmony □□□□□; contrast emphasis □□□□□.

Study points

- Compare two sheets with different line weights; use one dominant line family and one supporting family; maintain clean intersections; describe the intended visual effect before drawing.

Engineering / workplace application: Line hierarchy helps distinguish walls, openings, furniture, movement paths, and annotation in architectural drawings.

Illustrative example: Draw the same simple rectangle three ways: with uniform lines, with a heavier outer edge, and with a broken internal guide. Record how readability changes.

Common mistake: Using the same heavy line for every object removes hierarchy and makes the visual message ambiguous.

M1.03 — Practice exercises with lines (2 hours)

Concept and explanation

Line practice is a motor skill as well as a visual exercise. Work from light construction lines to confident final lines, keeping spacing, direction, and pressure consistent. Exercises may include parallel lines, converging lines, repeated curves, zig-zag patterns, and spiral sequences. The purpose is control and observation, not filling a page without analysis.

> **Malayalam support:** Line practice □□□□□□□□□□ hand control □□□□□□□□□□□□□□□□ exercise □□□. □□□□□ light construction lines □□□□□□□□□□ □□□□□□ confident final lines □□□□□.

Study points

- Use a consistent tool and sheet; repeat a pattern at least three times; compare the first and last attempt; annotate problems such as wobble, uneven spacing, or excessive pressure.

> **Engineering / workplace application:** Controlled linework supports quick site sketches, details, section indications, and visual notes made during architectural observation.

> **Illustrative example:** Create a practice sheet with five rows: parallel verticals, horizontals, diagonals, curves, and spirals. Rate each row for consistency after completion.

> **Common mistake:** Rushing to darken every line before the construction is correct makes correction difficult and produces muddy work.

M1.04 — Represent design principles with lines (2 hours)

Concept and explanation

The syllabus connects linework to rhythm, harmony, character, balance, and emphasis, and asks students to interpret scale and proportion. Rhythm may arise from repeated or progressively changing lines. Balance concerns visual weight. Emphasis directs attention to a selected region. Scale compares an object with a known reference, while proportion compares parts with one another. A successful composition makes these relationships visible rather than merely naming them.

Malayalam support: Rhythm, harmony, balance, emphasis □□□□□□ line arrangement □□□ □□□□□□□□□□. Scale □□□ known reference-□□□□□□□ □□□□□□□□□□□□□□; proportion parts □□□□□□□□□□ □□□□□□□□.

Study points

- Use a simple grid before arranging lines; create one dominant area and supporting areas; include a human figure or known object when showing scale; check whether repeated elements are evenly spaced.

Engineering / workplace application: Architectural façades, paving patterns, screens, stair rhythms, and circulation diagrams use these principles to organise visual information.

Illustrative example: Design a strip using repeated vertical lines, one contrasting diagonal group, and a human figure symbol. Explain the rhythm, emphasis, and scale decisions.

Common mistake: Calling every repeated pattern “harmony” without explaining spacing, similarity, or contrast weakens the design analysis.

Module summary: Line classification becomes useful when controlled linework is connected to visual hierarchy and architectural communication.

OFFICIAL MODULE 2

Graphical Language, Visual Composition and Graphic Conventions

Architectural graphical language combines points, lines, shapes, tone, texture, figure and ground, lettering, symbols, scale, and media-based studies. Students move from isolated elements to organised visual compositions and apply shading and graphic conventions in a disciplined way.

Hours: 12

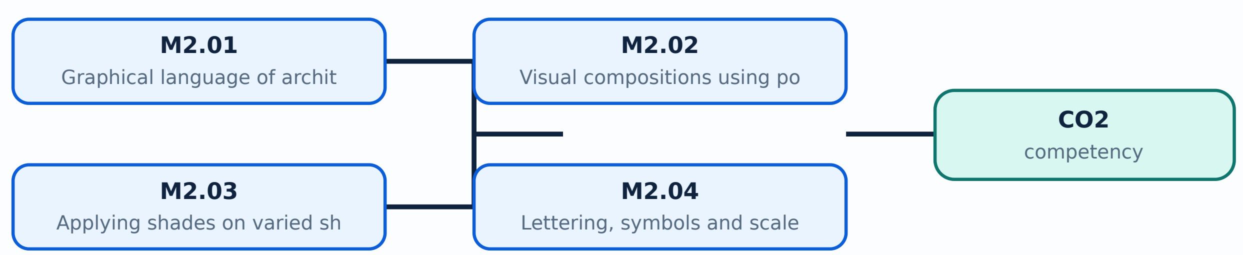
CO2 · Bloom level: Applying

Learning goals

- Explain how points, lines, shapes, tone, texture, figure, and ground contribute to a composition.
- Construct visual studies using varied shapes and media such as pencil, charcoal, and pastel.
- Apply lettering, symbols, scale, logo design, collage, and calligraphy as graphical communication tools.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-2: the listed topics build the module competency.

M2.01 — Graphical language of architecture (2 hours)

Concept and explanation

Graphical language is the system by which architectural ideas are communicated visually. It includes the selected marks, symbols, shapes, tones, textures, scale conventions, annotations, and page organisation used to make an idea understandable. A graphical language should be consistent: similar objects should be represented in similar ways and the viewer should be able to identify hierarchy and intent.

Malayalam support: Graphical language □□□□□□ architecture idea communicate □□□□□□□□□□□ visual system □□□. Symbols, line weight, scale, annotation □□□□□□ consistent □□□□□□□□□□.

Study points

- Start with the message before choosing a mark; use a legend when symbols may be unfamiliar; keep text, symbols, and visual elements aligned; distinguish observation from interpretation.

Engineering / workplace application: A site analysis board, floor-plan presentation, concept diagram, and portfolio page each use a graphical language appropriate to their purpose.

Illustrative example: Make a small legend for a room-study sheet containing symbols for movement, light, vegetation, and built elements, then apply it consistently.

Common mistake: Decorating a sheet without considering what the viewer must understand produces a visually busy but ineffective composition.

M2.02 — Visual compositions using points, lines, shapes, tone and texture (4 hours)

Concept and explanation

A visual composition arranges elements in a field so that the whole communicates more than isolated marks. Points can indicate focus or location; lines guide movement; shapes create masses; tone establishes light and dark relationships; texture communicates surface character. Figure is the active form or subject, while ground is the surrounding field. Varying size, position, repetition, and contrast changes the visual balance.

Malayalam support: Point, line, shape, tone, texture □□□□□□ composition-□ □□□□□□□□□□ roles □□□□□□□□□□□. Figure subject/active form □□□; ground surrounding field □□□.

Study points

- Use a limited number of elements first; establish a figure-ground relationship; check balance from a distance; preserve negative space; compare a tonal study in pencil with one in charcoal or pastel.

Engineering / workplace application: Presentation boards, logo studies, façade concepts, and interior mood sheets depend on the organisation of figure, ground, tone, and texture.

Illustrative example: Create three thumbnail compositions: point-dominant, line-dominant, and shape-dominant. In each, identify the figure, ground, focus, and balance.

Common mistake: Adding many textures without a hierarchy can hide the main figure and reduce contrast.

M2.03 — Applying shades on varied shapes (2 hours)

Concept and explanation

Shading describes how light interacts with form. First identify the light direction, then separate light, half-tone, core shadow, and cast shadow as appropriate to the exercise. Cuboids, prismatic, spherical, and globular objects require different transitions: planar forms show clearer tonal changes, while rounded forms need gradual transitions. The supplied syllabus expects study of varied shapes in different media.

Malayalam support: Light direction □□□□□□□□□□□ light, half-tone, core shadow, cast shadow □□□□ tonal relationships □□□□□□□□□. Planar shapes-□ change sharper □□□□□□□□□□; rounded forms-□ gradual □□□□□□□□□□.

Study points

- Draw a light arrow; keep the darkest tone near the logic of the form; use hatching direction consistently; avoid outlining every shadow edge with an unrelated dark line.

Engineering / workplace application: Shading is used in product sketches, material studies, interior perspectives, and quick form exploration before detailed drafting.

Illustrative example: Shade a cuboid, sphere, and prism under the same upper-left light source. Label light, shade, and cast-shadow regions.

Common mistake: Placing the cast shadow in the wrong direction makes the light source physically inconsistent.

M2.04 — Lettering, symbols and scale (4 hours)

Concept and explanation

Lettering and symbols make a graphical sheet readable. Lettering should be legible, aligned, and consistent in height and spacing. Symbols simplify repeated or conventional information, while scale connects a drawing to actual dimensions. A scale statement such as 1:100 means one unit on the drawing represents one hundred corresponding units in reality; the drawing must state or imply the scale clearly.

Malayalam support: Lettering readable □□□□□□□□□□; symbols repeated information simplify □□□□□□□. Scale drawing size-□□□ actual size-□□□ □□□□ ratio □□□. 1:100 □□□□□□ drawing-□□ 1 unit = reality-□□□□ 100 units □□□□□□□.

Study points

- Practise uniform letter height; keep guide lines light; distinguish labels from dimensions; write the scale near the title; never change scale silently within one sheet.

Engineering / workplace application: Plans, elevations, diagrams, title blocks, signage, and presentation boards all depend on lettering and symbols for quick interpretation.

Illustrative example: At scale 1:100, a 3 m wall is represented by 30 mm. State the units and show the calculation: drawing length = actual length / scale factor.

Common mistake: A beautiful drawing with illegible labels is not an effective architectural communication.

M2.ST1 — Series Test I (2 hours)

Concept and explanation

The official outline lists Series Test I under Module II. Prepare by revising graphical language, points/lines/shapes, tone and texture, figure-ground, varied-shape studies, lettering, symbols, scale, logo design, collage, and calligraphy. This is an official assessment item, not an additional drawing topic.

Malayalam support: Series Test I-□□□□□□ Module II-□□ graphical language, composition, shading, lettering, symbols, scale □□□□□□ revise □□□□□□□. □□□ official assessment item □□□.

Study points

- Practise one short composition and one scale/lettering task under time control; keep the sheet labelled; review line quality and presentation before submission.

Engineering / workplace application: A timed study trains the same visual decision-making required in studio submissions.

Illustrative example: Make a revision card containing the definitions of figure-ground, tone, texture, symbol, and scale, followed by one small illustrative sketch.

Common mistake: Beginning a timed sheet without a layout plan can leave insufficient space for labels or the final composition.

Module summary: Points, shapes, tones, textures, lettering, symbols, and scale form a coherent graphical language when their hierarchy is planned.

OFFICIAL MODULE 3

Free-Hand Sketching of Indoor and Outdoor Spaces

Free-hand sketching develops observation, proportion, spatial understanding, and visual communication without relying on a fully constructed instrument drawing. The syllabus includes types of three-dimensional views, indoor and outdoor sketching, still life, interiors, exteriors, and drawing from observation.

Hours: 15

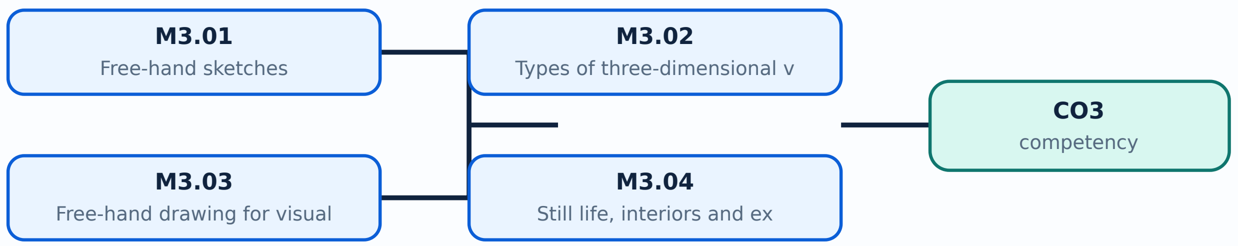
CO3 · Bloom level: Applying

Learning goals

- Describe the purpose and basic differences of common three-dimensional views.
- Use observation to place major masses, axes, proportions, and details in a free-hand sketch.
- Produce pencil sketches of still life, interiors, and exteriors appropriate to visual and architectural representation.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-3: the listed topics build the module competency.

M3.01 — Free-hand sketches (1 hours)

Concept and explanation

A free-hand sketch is a quick, observed drawing made without relying on precise drafting instruments. It is not careless drawing: the student still checks proportion, alignment, perspective cues, negative space, and the relationship between major and minor forms. Construction lines may remain light while the final accents are selective.

Malayalam support: Free-hand sketch □□□□□□ instruments □□□□□□□ observation □□□□□□□□□□□□□□ □□□□□□□□□ quick drawing □□□. Quick □□□□□□ proportion, alignment, perspective cues □□□□□□ □□□□□□□□□□□□.

Study points

- Begin with a bounding box or axis; establish large forms before details; keep the pencil loose; use selective dark accents; write the observation time or subject when maintaining a sketch record.

Engineering / workplace application: Architects use free-hand sketches during site visits, client discussions, early concept development, and design reviews.

Illustrative example: Observe a simple chair for five minutes. Draw its bounding box, main axes, large masses, and only then add two characteristic details.

Common mistake: Starting with small details such as windows or leaves before placing the overall mass usually produces proportion errors.

M3.02 — Types of three-dimensional views (3 hours)

Concept and explanation

Three-dimensional views communicate depth and form. A pictorial view may use parallel projection, such as an isometric or oblique view, or perspective cues in which receding edges appear to converge toward vanishing points. The syllabus requires students to discuss types of three-dimensional views; the practical choice should match the communication need and the level of accuracy required.

Malayalam support: 3D view-□ depth □□□□□□□□□□. Isometric/oblique parallel projection □□□□□□□□□□□; perspective-□ receding lines vanishing point-□□□□□□ converge □□□□□□□.

Study points

- Identify the horizon/eye level where relevant; keep parallel systems consistent; use a simple box before adding complex forms; do not mix incompatible projection rules in one sketch.

Engineering / workplace application: Interior views, product studies, furniture, and building massing use different pictorial choices depending on whether measurement or visual realism is more important.

Illustrative example: Draw the same cube as an oblique study and as a one-point perspective study. Label the receding edges and explain the visual difference.

Common mistake: Drawing one face in isometric and another in unplanned perspective makes the object unstable.

M3.03 — Free-hand drawing for visual and architectural representation (5 hours)

Concept and explanation

Architectural free-hand drawing combines observation with design communication. Establish the envelope, horizon or principal axes, proportion, major openings, and foreground/background relationship. Use line weight to separate near and far elements and use tone sparingly to show depth. The final sketch should communicate a place or idea even if it is not a measured drawing.

Malayalam support: Architectural sketch-□ envelope, principal axes, proportion, openings, foreground/background □□□□□□ □□□□□ □□□□□□□□□□□. Line weight-□□ tone-□□ depth □□□□□□□□□ □□□□□□□□□□□.

Study points

- Use a three-stage sequence: construction, clarification, selective rendering; compare the sketch with the subject; keep the viewpoint stable; use people or furniture only when they clarify scale.

Engineering / workplace application: Concept sketches help an architect test massing, circulation, façade rhythm, daylight ideas, and spatial atmosphere before detailed drafting.

Illustrative example: Select a doorway or small façade. Make three thumbnails from the same viewpoint, then develop the clearest one with light and dark hierarchy.

Common mistake: Over-rendering the sky or vegetation while leaving the building structure unclear reverses the communication priority.

M3.04 — Still life, interiors and exteriors in pencil (6 hours)

Concept and explanation

Still-life studies train proportion, material, and tonal observation. Interior sketches require attention to spatial enclosure, furniture, openings, and light direction. Exterior sketches require mass, site relation, horizon, and characteristic architectural elements. Pencil pressure, hatching, blending, and erasing should support the form rather than become decoration.

Malayalam support: Still life object proportions □□□□□□□□□□□□□; interior sketch enclosure, furniture, openings, light □□□□□□ □□□□□□□□□□; exterior sketch mass, site, horizon □□□□□□ □□□□□□□□□□.

Study points

- Choose a simple viewpoint; mark the eye level; place the largest shape first; reserve the whitest paper for highlights; use a consistent pencil sequence such as H for construction and softer grades for accents.

Engineering / workplace application: These studies support visual documentation of buildings, interior proposals, landscape edges, and material observations during design education.

Illustrative example: Complete a pencil study in three passes: large masses and perspective cues, secondary elements, and selective tonal accents. Add a short note describing the light source.

Common mistake: Changing the viewpoint halfway through a sketch causes furniture, openings, and horizon lines to disagree.

Module summary: Observation, spatial structure, and controlled pencil tone allow free-hand sketches to communicate indoor and outdoor architectural space.

OFFICIAL MODULE 4

Colour Theory and Visual Composition

Colour work connects visual perception with design decisions. Students study chromatic values, the colour wheel, primary/secondary/tertiary colours, colour charts, and schemes based on harmony, contrast, and degree of chromatism, then apply them to architectural forms and spaces.

Hours: 8

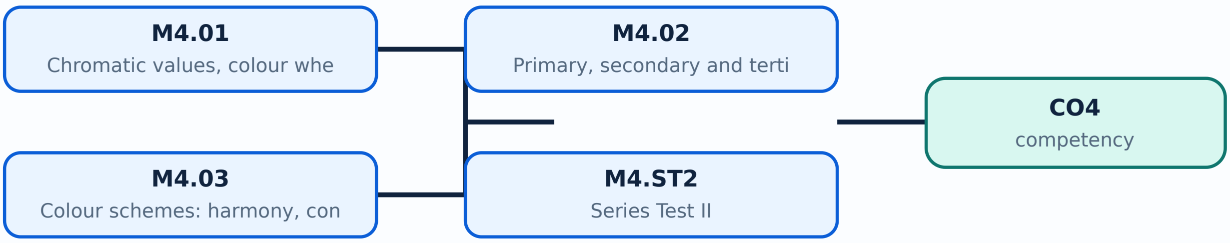
CO4 · Bloom level: Applying

Learning goals

- Explain chromatic value and the purpose of a colour wheel.
- Identify primary, secondary, and tertiary colours and prepare a colour wheel.
- Select and justify colour schemes using harmony, contrast, and degree of chromatism.

Module study tip

Read every topic in sequence, reproduce its example or procedure, and write a short explanation before attempting the module questions.



Learning map for Module module-4: the listed topics build the module competency.

M4.01 — Chromatic values, colour wheel and colour theory **(1 hours)**

Concept and explanation

Colour theory organises hue, value, and saturation/chromatism. Hue identifies the colour family, value indicates lightness or darkness, and chromatism describes the intensity or purity of a colour. A colour wheel arranges related hues so that relationships such as adjacent and opposing colours can be observed. The syllabus uses chromatic values and colour wheel as foundations for composition.

Malayalam support: Hue colour family □□□; value lightness/darkness □□□; chromatism □□□□□□□□□□ saturation intensity □□□. Colour wheel hue relationships □□□□□□□□□□□□ □□□□□□□□□□.

Study points

- Separate hue from value in your notes; make swatches cleanly; keep the medium consistent during a comparison; observe how a colour changes against different grounds.

Engineering / workplace application: Colour decisions affect wayfinding, mood, perceived scale, emphasis, and material expression in architectural interiors and façades.

Illustrative example: Prepare three swatches of one hue: high value, middle value, and low value. Then prepare three swatches with similar value but different chromatism.

Common mistake: Calling a colour “dark” when the real difference is saturation confuses value with chromatism.

M4.02 — Primary, secondary and tertiary colours; prepare a colour wheel (1 hours)

Concept and explanation

Primary colours are the starting set in the selected colour model; secondary colours result from combining pairs of primaries, and tertiary colours lie between a primary and a neighbouring secondary colour. Because media and colour models differ, label the chosen medium clearly. A useful colour wheel should be evenly arranged, named, and cleanly painted or rendered.

Malayalam support: Primary colours-□ □□□□□□ secondary colours □□□□□□□□; primary-□□□ secondary-□□□ □□□□□ tertiary colours □□□□. □□□□□□□□□□□□ medium/model wheel-□ □□□□□□□□□□□□□□.

Study points

- Use a circular guide; mix small test quantities; label each sector; keep the order consistent; avoid presenting a medium-specific wheel as a universal physical law without explanation.

Engineering / workplace application: Colour wheels help select presentation palettes, interior accents, signage contrasts, and material combinations.

Illustrative example: Construct a twelve-sector wheel for the chosen medium, label primary/secondary/tertiary positions, and add a short note about the mixing method.

Common mistake: Uneven sectors or unlabelled mixtures make the wheel difficult to use for later analysis.

M4.03 — Colour schemes: harmony, contrast and degree of chromatism (6 hours)

Concept and explanation

A colour scheme is a deliberate relationship among colours in a composition. Harmony may be built from related hues or controlled values; contrast may arise from opposing hues, light-dark difference, warm-cool difference, or saturation difference. Degree of chromatism controls how vivid or muted the scheme appears. In architectural forms and spaces, colour should support use, orientation, atmosphere, material reading, and visual hierarchy.

Malayalam support: Harmony related colours-□□□□ controlled relationship □□□; contrast difference emphasize □□□□□□□□. Chromatism □□□□□□□□□□ vivid, □□□□□□□□ muted appearance □□□□□□□□.

Study points

- State the purpose of the space before selecting a scheme; test the palette in small samples; check contrast for legibility; keep a dominant, secondary, and accent relationship; note lighting conditions.

Engineering / workplace application: Hospitals, classrooms, public interiors, façades, and wayfinding systems require different balances of calm, emphasis, visibility, and identity.

Illustrative example: Create two palettes for the same room: a harmonious low-chromatism scheme and a high-contrast scheme. Explain the expected effect on focus, comfort, and wayfinding.

Common mistake: Selecting colours only because they are individually attractive can produce a discordant scheme when value and saturation are not controlled.

M4.ST2 — Series Test II (2 hours)

Concept and explanation

The official outline lists Series Test II under Module IV. Revise chromatic values, colour-wheel construction, primary/secondary/tertiary colours, colour charts, harmony, contrast, chromatism, and application to architectural forms and spaces. The test duration is recorded as 2 hours in the supplied PDF.

Malayalam support: Series Test II-□□□□□□ colour theory, wheel, primary/secondary/tertiary colours, harmony, contrast, chromatism □□□□□□ revise □□□□□□□. PDF-□ duration 2 hours □□□.

Study points

- Practise a labelled colour wheel and one written comparison of harmony versus contrast; prepare clean swatches and explain choices in English technical terminology.

Engineering / workplace application: Timed colour analysis improves the ability to justify a palette rather than merely name colours.

Illustrative example: Make a one-page revision sheet with a colour wheel, a harmony palette, a contrast palette, and three definitions: value, harmony, chromatism.

Common mistake: Submitting a colour sheet without labels or without explaining the relationship between colours weakens the answer.

Module summary: Colour theory becomes architectural design knowledge when colour relationships are tested against composition, space, light, and use.

Formula and Notation Bank

Formula note: The supplied syllabus is primarily procedural or conceptual and does not prescribe a compulsory numerical formula bank. Focus on the official terminology, sequences, records, and practical demonstrations listed in the modules.

Diagram Library for Rapid Revision

Module 2: Graphical

g-map diagram.

Module 3: Free-Hand

its labelled learning-map diagram.

Module 4: Colour Theory

Open the module to view its labelled learning-map diagram.

Official Activities and Practical Requirements

Official activities / self-learning

- The supplied PDF does not provide a separate self-learning activity list. Use the drawing exercises, visual compositions, lettering, sketching, and colour-wheel work listed under the official modules as guided practice.

Practical requirements / equipment

- No separate equipment list is supplied in the PDF. Any studio materials used for practice should follow the department’s approved drawing-lab requirements.

Assessment Methodology

Official assessment information is reproduced below from the supplied syllabus.

- The supplied PDF identifies a Series Test component of 4 hours in total.
- Series Test I is listed under Module II for 2 hours.
- Series Test II is listed under Module IV for 2 hours.
- The PDF does not specify CIE/ESE marks or a separate end-semester examination pattern; this lesson does not invent one.

Module-wise Questions and Answers

module-1 — Graphic Representation and Lines

1. Explain Classify types of lines. **[3 marks]**

Lines are the primary marks used in visual and architectural representation. The syllabus names vertical, horizontal, diagonal, zig-zag, curvilinear, and spiral lines. Each type has a directional and expressive character: vertical lines can suggest height or stability, horizontal lines calmness, diagonal lines movement, and curvilinear lines softness or continuity. Classification should be based on geometry and visual behaviour, not on decoration alone.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

2. Explain Characteristics and visual impacts of lines. **[3 marks]**

The visual impact of a line depends on direction, length, thickness, continuity, spacing, and repetition. A heavy continuous line may establish an edge, while a thin or broken line may indicate a secondary or hidden condition. In composition, repeated lines can create rhythm; related line qualities can create harmony; a contrasting line can create emphasis. These are visual relationships, so the student should observe them in actual sheets and sketches.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

3. Explain Practice exercises with lines. **[3 marks]**

Line practice is a motor skill as well as a visual exercise. Work from light construction lines to confident final lines, keeping spacing, direction, and pressure consistent. Exercises may include parallel lines, converging lines, repeated curves, zig-zag patterns, and spiral sequences. The purpose is control and observation, not filling a page without analysis.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

4. Explain Represent design principles with lines. **[3 marks]**

The syllabus connects linework to rhythm, harmony, character, balance, and emphasis, and asks students to interpret scale and proportion. Rhythm may arise from repeated or progressively changing lines. Balance concerns visual weight. Emphasis directs attention to a selected region. Scale compares an object with a known reference, while proportion compares parts with one another. A successful composition makes these relationships visible rather than merely naming them.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-2 — Graphical Language, Visual Composition and Graphic Conventions

5. Explain Graphical language of architecture. **[3 marks]**

Graphical language is the system by which architectural ideas are communicated visually. It includes the selected marks, symbols, shapes, tones, textures, scale conventions, annotations, and page organisation used to make an idea understandable. A graphical language should be consistent: similar objects should be represented in similar ways and the viewer should be able to identify hierarchy and intent.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

6. Explain Visual compositions using points, lines, shapes, tone and texture. **[3 marks]**

A visual composition arranges elements in a field so that the whole communicates more than isolated marks. Points can indicate focus or location; lines guide movement; shapes create masses; tone establishes light and dark relationships; texture communicates surface character. Figure is the active form or subject, while ground is the surrounding field. Varying size, position, repetition, and contrast changes the visual balance.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

7. Explain Applying shades on varied shapes. **[3 marks]**

Shading describes how light interacts with form. First identify the light direction, then separate light, half-tone, core shadow, and cast shadow as appropriate to the exercise. Cuboids, prismatic, spherical, and globular objects require different transitions: planar forms show clearer tonal changes, while rounded forms need gradual transitions. The supplied syllabus expects study of varied shapes in different media.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

8. Explain Lettering, symbols and scale. **[3 marks]**

Lettering and symbols make a graphical sheet readable. Lettering should be legible, aligned, and consistent in height and spacing. Symbols simplify repeated or conventional information, while scale connects a drawing to actual dimensions. A scale statement such as 1:100 means one unit on the drawing represents one hundred corresponding units in reality; the drawing must state or imply the scale clearly.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-3 — Free-Hand Sketching of Indoor and Outdoor Spaces

9. Explain Free-hand sketches. **[3 marks]**

A free-hand sketch is a quick, observed drawing made without relying on precise drafting instruments. It is not careless drawing: the student still checks proportion, alignment, perspective cues, negative space, and the relationship between major and minor forms. Construction lines may remain light while the final accents are selective.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

10. Explain Types of three-dimensional views. **[3 marks]**

Three-dimensional views communicate depth and form. A pictorial view may use parallel projection, such as an isometric or oblique view, or perspective cues in which receding edges appear to converge toward vanishing points. The syllabus requires students to discuss types of three-dimensional views; the practical choice should match the communication need and the level of accuracy required.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

11. Explain Free-hand drawing for visual and architectural representation. [3 marks]

Architectural free-hand drawing combines observation with design communication. Establish the envelope, horizon or principal axes, proportion, major openings, and foreground/background relationship. Use line weight to separate near and far elements and use tone sparingly to show depth. The final sketch should communicate a place or idea even if it is not a measured drawing.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

12. Explain Still life, interiors and exteriors in pencil. [3 marks]

Still-life studies train proportion, material, and tonal observation. Interior sketches require attention to spatial enclosure, furniture, openings, and light direction. Exterior sketches require mass, site relation, horizon, and characteristic architectural elements. Pencil pressure, hatching, blending, and erasing should support the form rather than become decoration.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

module-4 — Colour Theory and Visual Composition

13. Explain Chromatic values, colour wheel and colour theory. [3 marks]

Colour theory organises hue, value, and saturation/chromatism. Hue identifies the colour family, value indicates lightness or darkness, and chromatism describes the intensity or purity of a colour. A colour wheel arranges related hues so that relationships such as adjacent and opposing colours can be observed. The syllabus uses chromatic values and colour wheel as foundations for composition.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

14. Explain Primary, secondary and tertiary colours; prepare a colour wheel. **[3 marks]**

Primary colours are the starting set in the selected colour model; secondary colours result from combining pairs of primaries, and tertiary colours lie between a primary and a neighbouring secondary colour. Because media and colour models differ, label the chosen medium clearly. A useful colour wheel should be evenly arranged, named, and cleanly painted or rendered.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

15. Explain Colour schemes: harmony, contrast and degree of chromatism. **[3 marks]**

A colour scheme is a deliberate relationship among colours in a composition. Harmony may be built from related hues or controlled values; contrast may arise from opposing hues, light-dark difference, warm-cool difference, or saturation difference. Degree of chromatism controls how vivid or muted the scheme appears. In architectural forms and spaces, colour should support use, orientation, atmosphere, material reading, and visual hierarchy.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

16. Explain Series Test II. **[3 marks]**

The official outline lists Series Test II under Module IV. Revise chromatic values, colour-wheel construction, primary/secondary/tertiary colours, colour charts, harmony, contrast, chromatism, and application to architectural forms and spaces. The test duration is recorded as 2 hours in the supplied PDF.

Exam focus: State the definition or purpose, explain the working or sequence, and add one application or caution.

Practice Model Question Paper

Practice Model Paper — Not an Official Examination Paper

This paper is generated from the supplied module coverage for revision and does not claim to reproduce the official examination pattern.

1. **1.** Define or explain *Classify types of lines* with one relevant application. (5 marks)
2. **2.** Define or explain *Characteristics and visual impacts of lines* with one relevant application. (5 marks)
3. **3.** Define or explain *Graphical language of architecture* with one relevant application. (5 marks)
4. **4.** Define or explain *Visual compositions using points, lines, shapes, tone and texture* with one relevant application. (5 marks)
5. **5.** Define or explain *Free-hand sketches* with one relevant application. (5 marks)
6. **6.** Define or explain *Types of three-dimensional views* with one relevant application. (5 marks)
7. **7.** Define or explain *Chromatic values, colour wheel and colour theory* with one relevant application. (5 marks)
8. **8.** Define or explain *Primary, secondary and tertiary colours; prepare a colour wheel* with one relevant application. (5 marks)

Writing advice: Use headings, standard terminology, and labelled diagrams or procedures where relevant.

Quick Revision

Module-wise key points

- **Graphic Representation and Lines:** Line classification becomes useful when controlled linework is connected to visual hierarchy and architectural communication.
- **Graphical Language, Visual Composition and Graphic Conventions:** Points, shapes, tones, textures, lettering, symbols, and scale form a coherent graphical language when their hierarchy is planned.
- **Free-Hand Sketching of Indoor and Outdoor Spaces:** Observation, spatial structure, and controlled pencil tone allow free-hand sketches to communicate indoor and outdoor architectural space.

- **Colour Theory and Visual Composition:** Colour theory becomes architectural design knowledge when colour relationships are tested against composition, space, light, and use.

Exam checklist

- Write the official terminology before adding explanation.
- Use a labelled diagram or step sequence when it improves the answer.
- Check conditions, boundaries, safety, hygiene, and documentation points.
- Separate official syllabus information from your own example.

Last-minute revision: Review the coverage table, formula bank, diagram library, and common-mistake notes inside every topic.

Glossary

Technical glossary

Term	Short meaning
Classify types of lines	Lines are the primary marks used in visual and architectural representation. The syllabus names vertical, horizontal, diagonal, zig-zag, curvilinear, and spiral lines. Each type has a directional and expressive character: vertical lines can suggest height or s
Characteristics and visual impacts of lines	The visual impact of a line depends on direction, length, thickness, continuity, spacing, and repetition. A heavy continuous line may establish an edge, while a thin or broken line may indicate a secondary or hidden condition. In composition, repeated lines ca
Practice exercises with lines	Line practice is a motor skill as well as a visual exercise. Work from light construction lines to confident final lines, keeping spacing, direction, and pressure consistent. Exercises may include parallel lines, converging lines, repeated curves, zig-zag patt
Represent design principles with lines	The syllabus connects linework to rhythm, harmony, character, balance, and emphasis, and asks students to interpret scale and proportion. Rhythm may arise from repeated or progressively changing lines. Balance concerns visual weight. Emphasis directs attention
Graphical language of architecture	Graphical language is the system by which architectural ideas are communicated visually. It includes the selected marks, symbols, shapes, tones, textures, scale conventions, annotations, and page organisation used to make an idea understandable. A graphical la

Term	Short meaning
Visual compositions using points, lines, shapes, tone and texture	A visual composition arranges elements in a field so that the whole communicates more than isolated marks. Points can indicate focus or location; lines guide movement; shapes create masses; tone establishes light and dark relationships; texture communicates su
Applying shades on varied shapes	Shading describes how light interacts with form. First identify the light direction, then separate light, half-tone, core shadow, and cast shadow as appropriate to the exercise. Cuboids, prismatic, spherical, and globular objects require different transitions:
Lettering, symbols and scale	Lettering and symbols make a graphical sheet readable. Lettering should be legible, aligned, and consistent in height and spacing. Symbols simplify repeated or conventional information, while scale connects a drawing to actual dimensions. A scale statement suc
Series Test I	The official outline lists Series Test I under Module II. Prepare by revising graphical language, points/lines/shapes, tone and texture, figure-ground, varied-shape studies, lettering, symbols, scale, logo design, collage, and calligraphy. This is an official
Free-hand sketches	A free-hand sketch is a quick, observed drawing made without relying on precise drafting instruments. It is not careless drawing: the student still checks proportion, alignment, perspective cues, negative space, and the relationship between major and minor for
Types of three-dimensional views	Three-dimensional views communicate depth and form. A pictorial view may use parallel projection, such as an isometric or oblique view, or perspective cues in which receding edges appear to converge toward vanishing points. The syllabus requires students to di
Free-hand drawing for visual and architectural representation	Architectural free-hand drawing combines observation with design communication. Establish the envelope, horizon or principal axes, proportion, major openings, and foreground/background relationship. Use line weight to separate near and far elements and use ton
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Primary, secondary and tertiary colours; prepare a colour wheel	Primary colours are the starting set in the selected colour model; secondary colours result from combining pairs of primaries, and tertiary colours lie between a primary and a neighbouring secondary colour. Because media and colour models differ, label the cho
Colour schemes: harmony, contrast and degree of chromatism	A colour scheme is a deliberate relationship among colours in a composition. Harmony may be built from related hues or controlled values; contrast may arise from opposing hues, light-dark difference, warm-cool difference, or saturation difference. Degree of ch

Term	Short meaning
Series Test II	The official outline lists Series Test II under Module IV. Revise chromatic values, colour-wheel construction, primary/secondary/tertiary colours, colour charts, harmony, contrast, chromatism, and application to architectural forms and spaces. The test duratio

Official References and Resources

References listed in the supplied syllabus

1. Paul Laseau, Graphic Thinking for Architects and Designers, John Wiley & Sons, New York, 2001.
2. V. S. Pramar, Design Fundamentals in Architecture, Somaiya Publications Pvt. Ltd., New Delhi, 1973.
3. Wucius Wong, Principles of Two-Dimensional Design.
4. Francis D. K. Ching, Architecture: Form, Space and Order, Van Nostrand Reinhold Co., Canada, 1979.

In-page Search